

Mathematics for Biomedical Engineering

Coursework Assignment

To be submitted by Friday 7th November, 5pm

Each student must hand in their own answers for the problem sets, with this completed cover sheet as the first page.

Homework problems should be uploaded to KEATS as a single, legible PDF file by the stated deadline. If taking pictures of handwritten homework, post-processing software such as CamScanner, OfficeLens or equivalent should be used to improve legibility.

You must show your work, provide clear diagrams (when necessary) and all the necessary explanations. Discussion with peers and teamwork are encouraged, but submissions must be composed individually (such that each student should be able to explain their submission orally if asked).

Submissions that are hard to read or submitted in a file format different from PDF or as multiple files **will not be marked**.

DECLARATION BY STUDENT

- This assignment is entirely my own work.
- I understand what is meant by plagiarism and collusion and have signed at enrolment the declaration concerning the avoidance of plagiarism and collusion.
- I understand that plagiarism and collusion are serious academic offences that may result in disciplinary action being taken.
- I understand that I must submit work before the deadline and that failure to do so will result in capped marks.

NAME: _____

SIGNATURE: _____

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1. Given

$$x = \ln t + t, \quad y = \sin t,$$

calculate the following:

(a) $\frac{dy}{dx}$

[5 marks]

(b) $\frac{d^2y}{dx^2}$

[5 marks]

2. (i) Write down *both* the Cartesian and vector equations for the line passing through the point $P(1, 2, 3)$ with direction vector $(2, 3, 4)$.

[2 marks]

(ii) Calculate the angle between this line and the plane given by

$$x + y + 2z = 4 \quad (1)$$

What is the distance of the plane to the origin?

[3 marks]

(iii) Find the point of intersection of the line given in (i) and the plane given in (ii).

[3 marks]

(iv) A second plane is given by

$$x - y + z = 2 \quad (2)$$

What is the direction of the line of intersection of the two planes and why?

[2 marks]

3. (i) A matrix $\mathbf{A}$ has an eigen vector $\mathbf{x}$ if

$$\mathbf{Ax} = \lambda \mathbf{x} \quad (3)$$

where λ is the corresponding eigen value. Explain what is meant by an eigen vector and an eigen value in the context of the above equation. Starting from this equation, derive the characteristic eigen vector equation.

[3 marks]

(iii) Find the eigen values for the matrix given by

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{pmatrix} \quad (4)$$

[4 marks]

(iii) For one of the eigen values found above (the rational eigen value), derive the corresponding normalised eigen vector.

[3 marks]